Original Article

Deep learning-based no-reference quality assessment of anterior segment ultrasound biomicroscopy panoramic images

Qing-Hao Miao[1], Xiao-Chun Wang[1], Jun Yang[1], Xiao-Ning Wang[1], Xin-Qi Yu[1], You Zhou[1], Zhi-Yuan Zhao[1], Bin Wu[2,3], Sheng Zhou[1]

[1]State Key Laboratory of Advanced Medical Materials and Devices, Institute of Biomedical Engineering, Tianjin Institutes of Health Science, Chinese Academy of Medical Science and Peking Union Medical College, Tianjin, China; [2]Department of Visual Function Examination, Tianjin Eye Hospital, Tianjin, China; [3]Tianjin Key Lab of Ophthalmology and Visual Science, Tianjin, China

Contributions: (I) Conception and design: All authors; (II) Administrative support: XC Wang, S Zhou, B Wu, J Yang, XN Wang; (III) Provision of study materials or patients: B Wu; (IV) Collection and assembly of data: B Wu, XC Wang, S Zhou, QH Miao; (V) Data analysis and interpretation: QH Miao, XC Wang, S Zhou, B Wu, J Yang, XQ Yu, Y Zhou, ZY Zhao; (VI) Manuscript writing: All authors; (VII) Final approval of manuscript: All authors.

Correspondence to: Sheng Zhou, PhD. State Key Laboratory of Advanced Medical Materials and Devices, Institute of Biomedical Engineering, Tianjin Institutes of Health Science, Chinese Academy of Medical Science and Peking Union Medical College, Building 23, Huijin Cheng, East Tuanbo Avenue, North Beihua Road, Tuanbo New Town, Jinghai District, Tianjin 301600, China. Email: zhousheng@bme.pumc.edu.cn; Bin Wu, PhD. Department of Visual Function Examination, Tianjin Eye Hospital, No. 4 Gansu Road, Heping District, Tianjin 300020, China; Tianjin Key Lab of Ophthalmology and Visual Science, Tianjin, China. Email: tianjinyankewubin@163.com.

Background: Ultrasound biomicroscopy (UBM) enables high-resolution imaging of the anterior segment, essential for accurate diagnosis and anatomical assessment. However, the quality evaluation of UBM currently relies on subjective judgment, which is time-consuming and inconsistent. This study proposes a deep learning (DL)-based no-reference method for objective and automated UBM image quality assessment (IQA), facilitating reliable selection of high-quality images for clinical use.

Methods: A total of 1,154 clinical panoramic UBM images of the anterior segment were collected from Tianjin Eye Hospital. The YOLOv8s_DW_FOCUS model was employed to accurately extract the region of interest (ROI) and identify five key anatomical landmarks: the central corneal epithelium, central corneal endothelium, posterior lens capsule, left ciliary groove, and right ciliary groove. In collaboration with clinical ophthalmologists, eight key criteria for assessing anterior segment UBM image quality were established, integrating general medical image evaluation parameters and ophthalmic expertise. Based on these criteria, each frame was assigned a quality score. Images scoring 7 or higher were classified as high-quality, whereas those receiving a perfect score of 8 were considered standard. To validate the feasibility of our method, we conducted rigorous evaluations of its accuracy, focus on key regions, generalization capability, inter- and intra-class discrimination, and consistency in assessment results.

Results: The target detection model achieved a mean average precision (mAP) of 0.935, a recall of 0.898, and a precision of 0.925. Additionally, it effectively focused on key regions, as demonstrated by the heatmap analysis. The t-distributed stochastic neighbor embedding (t-SNE) plot further highlighted the model's strong discriminative capability across different classes and its excellent generalization performance. To assess the consistency between our no-reference quality assessment method and expert evaluations, we analyzed 174 standard images that had been subjectively selected by clinical ophthalmologists. Among them, 146 images received a score of 8, whereas 26 images scored 7, indicating a high level of agreement with clinical experts in identifying high-quality images. Moreover, our method applies stricter criteria for defining standard images, enabling a more precise selection of high-quality anterior segment UBM images.

Conclusions: The DL-based no-reference quality assessment method proposed in this study provides an

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objective evaluation of anterior segment UBM image quality. It effectively identifies high-quality images, significantly improving the efficiency of ophthalmic imaging professionals and demonstrating strong clinical potential for widespread adoption.

Keywords: Deep learning (DL); anterior segment ultrasound biomicroscopy images (anterior segment UBM images); object detection; reference-free quality assessment; medical image quality control

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Introduction

The anterior segment is a critical part of the eye, encompassing the region between the cornea and the lens. It includes key structures such as the cornea, sclera, anterior chamber, iris, pupil, and lens, all of which are essential for visual function and overall ocular health. Imaging the anterior segment provides valuable insights for diagnosing diseases, detecting lesions, and assessing anatomical structures and ocular parameters. Among various imaging techniques, ultrasound biomicroscopy (UBM) is particularly notable for its ability to capture high-resolution images of the anterior segment. Operating at ultrasound frequencies of 50–100 MHz, UBM enables detailed visualization and precise measurement of deep ocular structures, including the ciliary body, posterior chamber, and posterior lens capsule (1,2).

In ophthalmic clinical practice, medical image quality directly affects diagnostic accuracy and treatment effectiveness (3,4). Clinicians tend to prefer high-quality images to ensure diagnostic precision, with a primary focus on their practical value for diagnosis and treatment (5,6). Obtaining high-quality images that clearly capture the full structure and typical morphology of the eye is essential for precise diagnosis and successful treatment. However, ensuring the acquisition of high-quality images relies on accurate and reliable image quality assessment (IQA) (7).

Medical IQA is generally categorized into two main approaches: subjective and objective evaluation. Subjective evaluation relies on radiologists to assess image quality based on their clinical experience. Although widely used, this traditional method has several limitations: results depend entirely on the radiologist's expertise, leading to significant inter-observer variability; the process is labor-intensive and time-consuming; high workloads increase the likelihood of errors; and inexperienced doctors may struggle to obtain high-quality images in complex clinical settings (8,9). In contrast, objective evaluation employs quantitative algorithms or models to generate standardized image quality scores. It is typically classified into full-reference (FR-IQA), reduced-reference (RR-IQA), and no-reference (NR-IQA) methods. FR-IQA requires high-quality reference images and assesses image quality using metrics such as structural similarity (SSIM), root mean square error (RMSE), peak signal-to-noise ratio (PSNR), and visual information fidelity (VIF), which compare the test image to the reference image (10). RR-IQA evaluates key features extracted from both the reference and test images. NR-IQA, on the other hand, does not rely on a reference image and directly assesses image quality based on pixel-level features, offering efficiency, flexibility, and broad applicability. Compared to subjective evaluation, objective quality assessment eliminates inconsistencies and scoring biases associated with human judgment while maintaining a strong correlation with radiologists' subjective evaluations. As a result, objective IQA methods are increasingly being adopted in medical imaging applications (11-14).

Currently, there is no standardized system for assessing the quality of anterior segment UBM images, and automated quality evaluation remains underdeveloped. Subjective evaluation by clinicians remains the primary method, with objective assessment approaches still in their early stages. Several factors hinder the acquisition of standard UBM images that encompass all critical structures. These include the strict requirements for eye movement control and patient cooperation, anatomical structure overlap, and the need for real-time imaging. Additionally, individual anatomical variations and pathological differences among patients make it challenging to establish universal high-quality reference images. Consequently, traditional FR-IQA and RR-IQA methods may not be well-suited for UBM IQA, whereas NR-IQA is more appropriate. Clinicians assess image quality not only based on pixel-level attributes but also on the diagnostic value of the

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image content (15). Therefore, NR-IQA evaluation metrics should seamlessly integrate general quality parameters with structured features, efficiently incorporating prior clinical knowledge and statistical patterns to minimize false detection rates (16). Furthermore, in medical imaging, the goal is not to obtain the most aesthetically pleasing image, rather an interpretable image that contains all necessary diagnostic structures. Thus, NR-IQA should prioritize evaluating the quality of regions of interest (ROIs) that are most relevant to clinicians, rather than solely focusing on overall visual quality (17). This study is built upon these principles.

The rapid development of artificial intelligence (AI), particularly deep learning (DL) technologies, has significantly enhanced medical IQA by improving efficiency, automation, and precision. These advancements have revitalized image quality control, offering more reliable and scalable solutions. Piccini *et al.* proposed an automated IQA algorithm for evaluating cardiac magnetic resonance imaging (MRI) images, demonstrating a strong correlation with clinical assessments (18). Esses *et al.* developed a convolutional neural network (CNN)-based DL algorithm for screening non-diagnostic T2-weighted liver MRI images, achieving real-time performance with a low likelihood of misdiagnosis (19). Taye *et al.* explored the application of DL algorithms in trauma ultrasound IQA, providing valuable insights into automation and real-time quality control (20). Wang *et al.* introduced an ophthalmic optical coherence tomography (OCT) quality assessment system based on the ResNet-50 model, which automatically evaluates retinal OCT image quality based on signal occlusion, centralization, and the position of the ROI (21). Abramovich *et al.* developed a regression-based method for fundus IQA using the Inception-V3 model, demonstrating strong generalization capabilities (22). Taksoee-Vester *et al.* applied a U-Net-based model for standard plane recognition (8 planes), anatomical segmentation (28 features), and quality evaluation in fetal echocardiography, with the quality assessments of high-quality images closely matching those of clinical experts (23). Sujit *et al.* employed a deep convolutional neural network (DCNN) to automatically assess the quality of structural brain MRI images from multiple centers, achieving stable performance across institutions and devices (accuracy of 0.84), making it particularly suitable for large-scale screening and quality control of imaging data (24). To overcome the limitations of low image quality and limited computing power in mobile fundus imaging, Pérez *et al.* developed Mobile Fundus Quality Network (MFQ-Net), an ultra-lightweight CNN capable of real-time binary and three-level image quality classification. Despite its compact design, MFQ-Net achieves accuracy comparable to larger models, making it well-suited for primary screening and telemedicine applications (25). In summary, DL-based objective IQA methods have been widely applied in the medical field, but further refinement is needed to meet the demands of clinical practice. As AI technology continues to advance and medical image standardization progresses, DL is expected to play an increasingly critical role in medical image quality control.

This study addresses clinical needs by collaborating with ophthalmic imaging specialists to integrate general medical IQA parameters with domain-specific ophthalmic expertise. To enhance the quality evaluation of anterior segment UBM panoramic images, we have defined eight key quality metrics: Image Clarity (IC), Cornea Visibility (CV), Posterior Capsule Visibility (PCV), Posterior Capsule Horizontal Centering (PCHC), Anterior Vertical Alignment (AVA), Ciliary Groove Visibility (CGV), Ciliary Groove Focal Zone (CGFZ), and Ciliary Groove Horizontal Alignment (CGHA). Leveraging the YOLOv8s_DW_FOCUS DL model, we achieved high detection accuracy, effectively capturing fine anatomical details within anterior segment images. These quality indicators were then employed for NR-IQA, providing an objective and precise evaluation of UBM images. This method is designed to assist clinicians in efficiently extracting standard UBM cross-sectional views of the anterior segment and reliably identifying high-quality images. The results demonstrate that this NR-IQA approach closely aligns with subjective evaluations performed by clinicians, confirming its feasibility and strong potential for clinical application. We present this article in accordance with the CLEAR reporting checklist (available at https://qims.amegroups.com/article/view/10.21037/qims-2025-592/rc).

Methods

Datasets

The dataset for this study was sourced from patients who underwent panoramic UBM examinations at Tianjin Eye Hospital between 11 April 2019 and 1 December 2024. The images, captured by experienced ophthalmic imaging specialists, have a resolution of 1024×576 pixels. A total of 1,154 images were used for both training and

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testing, with the data split in an 80:20 ratio. The UBM device employed in this study was the MD-300L (MEDA Co., Ltd., Tianjin, China), featuring an ultrasound probe frequency of 50 MHz, a scan depth of 11 mm, and a width of 17.5 mm. To protect patient privacy and ensure data security, all UBM images were anonymized. The study was conducted in accordance with the Declaration of Helsinki and its subsequent amendments and received approval from the Medical Ethics Committee of Tianjin Eye Hospital (Approval No. KY-202417). As this was a retrospective study using anonymized UBM images, informed consent was not required.

To reflect real-world clinical conditions, the dataset consisted of a 2:1 mix of high-quality and low-quality UBM images, with image quality subjectively assessed by an ophthalmic imaging physician with 10 years of experience. The images were annotated independently by the physician using LabelMe (Massachusetts Institute of Technology, Cambridge, MA, USA) software, followed by a secondary review to ensure annotation accuracy. Each UBM image was annotated based on eight key quality control indicators, specifying the coordinates of five key points: the central corneal epithelium, central corneal endothelium, left and right ciliary grooves, and the posterior lens capsule. Additionally, rectangular bounding boxes were drawn around the ROIs. These annotations served as the gold standard for the object detection task.

Model

After a comprehensive comparison and validation of the accuracy and real-time performance of various classic object detection models, and taking into account the need for deployment on low-resource hardware, this study ultimately selected the YOLOv8s_DW_FOCUS model for key point and ROI detection. After detecting the objects in each frame, the model integrates the results into quality control indicators, enabling the calculation of a quality score for the current frame.

The YOLOv8 architecture consists of three primary components: the Backbone, the Neck, and the YOLO Head. The Backbone is responsible for extracting high-level features from the input image, consisting of multiple convolutional layers and network modules. The Neck facilitates the fusion and transmission of the features extracted by the Backbone, transforming them into a format suitable for object detection. Finally, the YOLO Head converts these processed feature maps into specific object detection outputs.

The network structure of the YOLOv8s_DW_FOCUS model is shown in *Figure 1*.

In this model, depthwise separable convolutions (DWConv) are used in place of standard convolutional layers. This replacement enables the use of more convolutional layers without significantly increasing computational costs, thereby deepening the network and enhancing its ability to capture small objects and intricate details. This is particularly advantageous for accurately detecting subtle key points in anterior segment UBM images.

Additionally, the FOCUS module is integrated into the input layer of the backbone. By reducing the spatial resolution of the input image and focusing on specific ROIs, this module enhances the efficiency of information extraction while filtering out unnecessary background information. As a result, the YOLOv8s_DW_FOCUS model significantly improves detection accuracy, especially in the detection of small objects and the precise localization of key points in anterior segment images.

Quality control indicators

By integrating general medical IQA parameters with clinical expertise in ophthalmology, this study effectively addresses the key clinical factors influencing the quality of anterior segment UBM images, while also exploring the feasibility of implementing quality control methods. In collaboration with imaging specialists from Tianjin Eye Hospital, eight essential quality control indicators were established, as summarized in *Table 1*.

To emphasize the most informative regions in anterior segment images, we extracted the ROI and evaluated image clarity by calculating the Laplacian variance of the ROI. The calculation process is described as follows:

Firstly, the pixel transformation values within the ROI are obtained using the following equation:

$$L(x,y)=\nabla^2 I(x,y)=\frac{\partial^2 I}{\partial x^2}+\frac{\partial^2 I}{\partial y^2} \quad [1]$$

Here, x and y denote the horizontal and vertical coordinates of a pixel in the ROI image, respectively. $I(x,y)$ represents the grayscale intensity at that location, $\nabla^2 I(x,y)$ denotes the Laplacian-transformed value at (x,y), and $L(x,y)$ refers to the pixel transformation value after applying the Laplacian operator.

Subsequently, the mean of the pixel transformation

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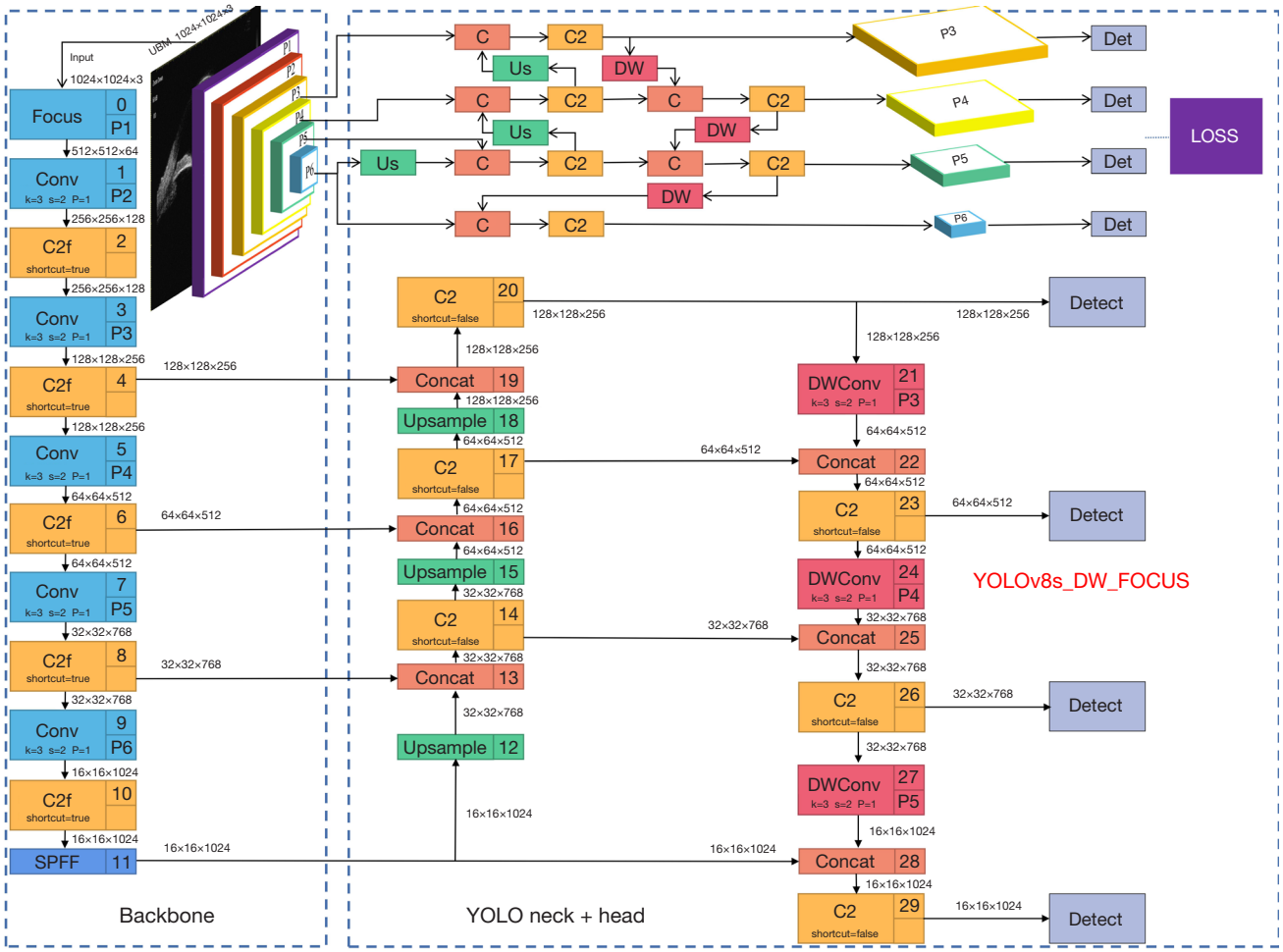


Figure 1 Network architecture of YOLOv8s_DW_FOCUS.

values is calculated using the following equation:

$$\mu = \frac{1}{M \times N} \sum_{x=1}^{M} \sum_{y=1}^{N} L(x, y) \quad [2]$$

Where μ represents the mean of the pixel transformation values, and $M \times N$ denotes the total number of pixels in the image.

Finally, the Laplacian variance *Var(L)* within the ROI is obtained.

$$Var(L) = \frac{1}{M \times N} \sum_{x=1}^{M} \sum_{y=1}^{N} \left(L(x, y) - \mu \right)^2 \quad [3]$$

High-quality images typically exhibit sharper edges and larger variations in grayscale gradients, leading to higher Laplacian variance values. In the early stages of the study, we empirically tuned the Laplacian variance threshold and compared the AI model's IC assessments with those of clinical experts to evaluate accuracy and consistency. To determine the optimal threshold, a set of UBM images with varying levels of clarity was selected. The Laplacian variance within the ROI of each image was computed, and the threshold was iteratively adjusted. The value that best differentiated image quality and showed the highest agreement with expert evaluations was ultimately chosen. Accordingly, the optimal threshold for *Var(L)* was set to 150.

The CV indicator requires the anterior and posterior surfaces of the cornea to display a crescent-shaped region of high echogenicity. A well-defined crescent structure reflects a uniform distribution of echo signals, ensuring the image's diagnostic value. Similarly, the PCV indicator requires the posterior lens capsule to exhibit a reverse crescent-shaped pattern of high echogenicity. This distinct reverse crescent structure reflects the optimal alignment of ultrasound waves

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Table 1 Key quality control indicators and descriptions

Quality control indicators	Abbreviation	Indicator specification
Image Clarity	IC	The Laplacian variance reflecting the clarity of the ROI image is greater than the threshold
Cornea Visibility	CV	Both the anterior and posterior surfaces of the cornea exhibit crescent-shaped echoes, with the average pixel brightness greater than the threshold
Posterior Capsule Visibility	PCV	The posterior capsule of the lens exhibits a reverse crescent-shaped echo, with the average pixel brightness greater than the threshold
Posterior Capsule Horizontally Centered	PCHC	The center of the posterior capsule of the lens is located within a ±1 mm ranges from the center of the image
Ciliary Groove Visibility	CGV	Both the left and right ciliary grooves are identified
Ciliary Groove Focal Zone	CGFZ	Both the left and right ciliary grooves are within the focal zone of the UBM imaging
Ciliary Groove Horizontal Alignment	CGHA	The inclination of the left and right ciliary grooves in the vertical direction is less than 5°
Anterior Vertical Alignment	AVA	The deviation between the center of the posterior capsule and the midpoint of the corneal anterior edge in the horizontal direction is less than 10°

ROI, region of interest; UBM, ultrasound biomicroscopy.

with the lens and the meridian section of the eye, which are essential for selecting high-quality images. The visibility of these anatomical structures was assessed by calculating the average pixel brightness values for the central corneal epithelium, central corneal endothelium, and posterior capsule regions. Following a similar approach to the threshold selection for *Var(L)*, a set of UBM images with varying brightness levels in the target regions was selected. The average pixel brightness values were calculated, and the threshold was iteratively adjusted. The AI-based assessments were then compared with expert clinical evaluations to verify consistency and discriminative effectiveness. As a result, the optimal average brightness threshold for both the CV and PCV was determined to be 27 (8-bit).

The actual dimensions of the UBM image are 17.5×11 mm. To evaluate horizontal alignment, the position of the posterior capsule center was analyzed. An image is considered horizontally centered if the horizontal coordinate of the posterior capsule lies within ±1 mm of the image center, increasing the likelihood of fully capturing the anterior segment structure. This criterion was used for the PCHC indicator.

In comparison to anterior segment optical coherence tomography (AS-OCT), UBM offers a significant advantage as ultrasound can penetrate the iris pigment epithelium, clearly displaying the ciliary body structure (26). In anterior segment UBM imaging, a single-element high-frequency ultrasound transducer generates a fixed focal region. Within this focal zone, the image maintains high resolution, but resolution diminishes sharply outside this area. For diagnostic reliability, the ciliary groove must be positioned within the focal zone. The MD-300L device has a focal zone located vertically between 5 and 7 mm in the image. The CGFZ indicator evaluates the vertical positioning of the ciliary groove, whereas the CGV indicator assesses its overall visibility.

To ensure proper alignment of the anterior segment structures, two additional indicators were established: The CGHA and AVA. These indicators measure the degree of tilt in the horizontal and vertical structures of the image. Testing and validation determined the optimal tilt angle thresholds for CGHA and AVA to be 5° and 10°, respectively.

In this study, each of the eight quality control indicators was formally assigned an equal weight of 1 point. However, during the design process, the clinical significance of each anatomical structure in the evaluation of UBM image quality was fully considered to improve the accuracy and clinical applicability of the assessment. For example, the PCV indicator is used to determine whether the scanning plane is close to the lens axis and the meridional section of the eye, with the posterior capsule playing a critical role in image quality evaluation. Therefore, the scores for the PCHC and AVA indicators, which are closely related to the

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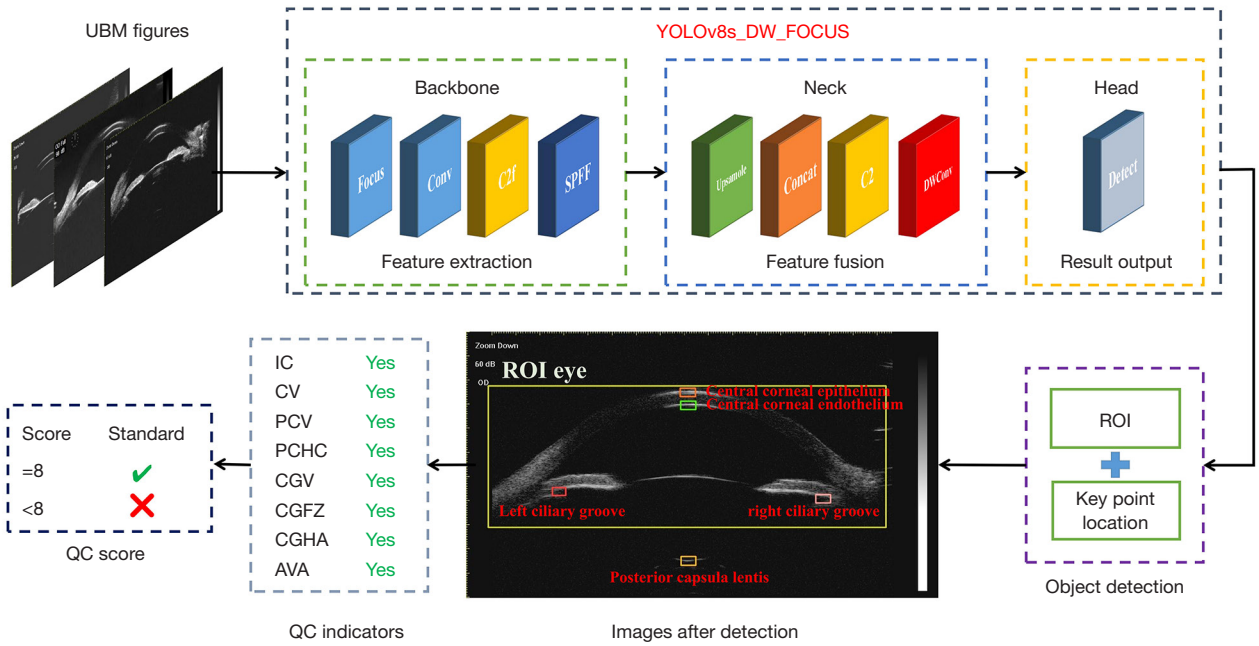


Figure 2 Block diagram of the overall process. AVA, anterior vertical alignment; CGFZ, ciliary groove focal zone; CGHA, ciliary groove horizontal alignment; CGV, ciliary groove visibility; CV, cornea visibility; IC, image clarity; PCHC, posterior capsule horizontal centering; PCV, posterior capsule visibility; QC, quality control; ROI, region of interest; UBM, ultrasound biomicroscopy.

posterior capsule, depend on the PCV assessment results, implicitly assigning greater weight to the posterior capsule in the scoring system. Similar dependency exists among the CGV, CGFZ, and CGHA indicators. Thus, although the scores are equally weighted, the design of the indicators inherently reflects differences in the clinical importance of the anatomical structures, balancing model simplicity with clinical relevance.

The quality score for each frame is displayed, with images scoring 7 or above classified as high-quality, and those achieving a perfect score of 8 designated as standard-plane images. In this study, a standard-plane image refers to a UBM image acquired along the meridional plane that passes through the visual axis, clearly depicting key anterior segment structures, including the central corneal epithelium and endothelium, the posterior lens capsule, and both the left and right ciliary grooves.

To be considered a standard-plane image, the frame must meet all eight predefined quality control criteria, which assess sharpness (IC), structural visibility (CV, PCV, CGV), positional accuracy (PCHC, CGFZ), and geometric alignment (CGHA, AVA). These criteria collectively ensure consistency in scanning planes and reliable anatomical representation.

Experimental procedure

This study employed DL techniques to automatically identify the ROI and five key points in anterior segment UBM images, while simultaneously evaluating image quality based on eight predefined quality control indicators. The primary objective of this approach is to assist clinicians in efficiently extracting standard anterior segment UBM cross-sectional views and selecting high-quality images. The workflow of this process is illustrated in *Figure 2*.

Initially, using 1,154 UBM images annotated by clinical ophthalmologists, the YOLOv8s_DW_FOCUS DL model performs real-time object detection to identify the ROI and the positions of five key points for each frame. The ROI detection box represents the actual size of the anterior segment; each key point is enclosed by a detection box measuring 30.72×17.28 pixels, with the center of the box serving as the key point's coordinate. After object detection, the necessary pixel-level and geometric information is extracted from the image. The frame is then evaluated based on the previously established eight quality control

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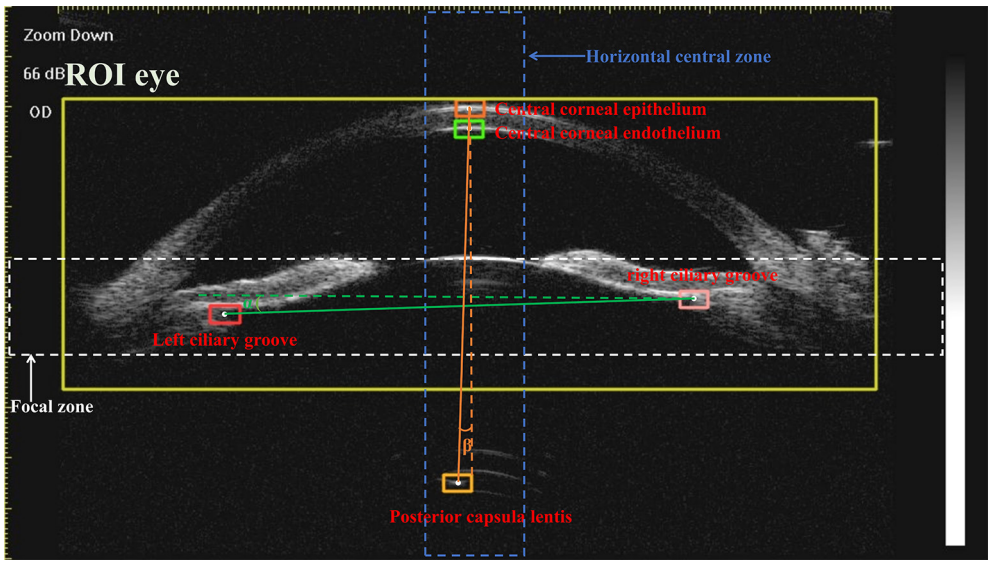


Figure 3 Parameter information required for quality control. ROI, region of interest.

criteria, with the parameters required for scoring displayed in *Figure 3*.

The IC indicator assesses the clarity of the image within the ROI by calculating the Laplacian variance. A score of 1 is assigned if the variance exceeds 150. The CV indicator requires the detection of both the central corneal epithelium and central corneal endothelium, with the average pixel intensity within their detection boxes exceeding 27 (8-bit). Similarly, the PCV indicator mandates the detection of the posterior lens capsule, with an average pixel intensity above 27. The PCHC indicator ensures that the posterior lens capsule is positioned within the horizontal central zone, defined as a ±1 mm range from the image's horizontal center. The CGV indicator requires the detection of both the left and right ciliary grooves in the image. For the CGFZ indicator, the left and right ciliary grooves must lie within the focal zone, defined as the vertical range between 5 and 7 mm in the image. The CGHA indicator ensures that the horizontal angle (α) between the left and right ciliary grooves is less than 5 degrees, whereas the AVA indicator ensures that the vertical angle (β) between the posterior lens capsule and the central corneal epithelium is less than 10 degrees.

Once these criteria are evaluated, the quality control score for each frame is displayed in real-time in the upper-right corner of the image, providing clinicians with an intuitive reference. Images that achieve a score of 7 or higher are classified as high-quality, whereas those receiving a perfect score of 8 are considered standard-plane images.

To ensure consistency between the proposed no-reference quality assessment model and the subjective evaluations of clinical experts, we selected 174 images previously identified as standard anterior segment UBM images by clinicians. These images were processed through the no-reference quality assessment model, and the results were compared to the clinicians' subjective evaluations to assess any discrepancies between the two approaches.

Environment setup

In this study, the YOLOv8s_DW_FOCUS model was employed, with the model training framework set to PyTorch 2.4.1 (Linux Foundation, San Francisco, CA, USA), Python version 3.11.9 (Python Software Foundation, Wilmington, DE, USA), and the GPU being an NVIDIA RTX 3090Ti 24 GB (NVIDIA, Santa Clara, CA, USA). The computation platform used was CUDA version 12.1 (NVIDIA).

Evaluation metrics

In this study, the performance of the object detection model was evaluated using several metrics, including mean average precision (mAP), precision, recall, and detection speed (FPS, frames per second). These metrics were used to compare the DL detection results with the ground truth.

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Table 2 Performance comparison of object detection models

Models	mAP	Recall	Precision	FPS
YOLOv5s	0.916	0.876	0.902	42.83
YOLOv8s	0.923	0.891	0.898	44.71
YOLOv8s_swin	0.923	0.898	0.910	43.64
YOLOv8s_SEAattention	0.920	0.893	0.905	44.25
YOLOv8s_GE	0.912	0.884	0.894	42.42
YOLOv8s_focus	0.923	0.889	0.902	44.64
YOLOv8s_DW_FOCUS	0.935	0.898	0.925	41.44
YOLOv8s_small	0.918	0.883	0.893	42.68
YOLOv8s-efficientViT	0.924	0.894	0.894	23.62
RT-DETR	0.898	0.864	0.852	18.10

FPS, frames per second; mAP, mean average precision.

To visually present discrepancies between the predicted results and the ground truth labels, a confusion matrix was employed. This approach allows for a clearer assessment of detection accuracy across different categories. Additionally, the precision-recall (P-R) curve was used to further analyze the model's classification and detection performance. The P-R curve illustrates the relationship between precision and recall at an Intersection over union (IoU) threshold of 0.5, offering a comprehensive evaluation of the model's overall performance.

Heatmaps were also employed to visualize the model's attention to different regions during image processing. The color intensity in the heatmaps indicates the degree of focus the model places on specific areas, with darker colors reflecting stronger attention. Finally, t-distributed stochastic neighbor embedding (t-SNE) was applied to map high-dimensional data into two- or three-dimensional space. This technique provides insights into how the model processes and differentiates between various categories of data, enabling a visual assessment of the training data quality (27).

Results

Initially, this study conducted a comprehensive analysis and evaluation of the overall performance of several commonly used object detection models for ROI detection and key point localization. The results of these tests are summarized in *Table 2*.

The results clearly demonstrate that the YOLOv8s_DW_FOCUS model outperformed other models in terms of mAP, recall, and precision. Its superior ability to detect small objects and capture fine details, while utilizing the same computational resources, makes it the most suitable model for ROI detection and keypoint localization in anterior segment UBM images. Additionally, the model achieved a detection speed of 41.44 FPS, significantly surpassing the clinical UBM detection requirement of 3–5 FPS. The confusion matrix and P-R curve for the YOLOv8s_DW_FOCUS model post-training are shown in *Figure 4*.

The confusion matrix shows that the YOLOv8s_DW_FOCUS model exhibited a low error rate in detecting ROIs and key points within the internal datasets, underscoring its robust classification capabilities. The P-R curve shows an area under the curve (AUPRC) of 0.935, indicating that the model effectively maintains high precision and recall, reflecting its exceptional performance.

Additionally, heatmaps provide an intuitive visualization of the neural network's feature maps. *Figure 5* illustrates the heatmaps and detection results for the target detection task in this study.

The YOLOv8s_DW_FOCUS model successfully identified the complete ROI and all key points in high-quality images. In contrast, for low-quality images, the model detected the ROI and three key points, with the missing key points also absent in the original images. In both high- and low-quality images, the global heatmap's darker regions were concentrated in the larger ROI area, whereas the local heatmap's darker regions were focused

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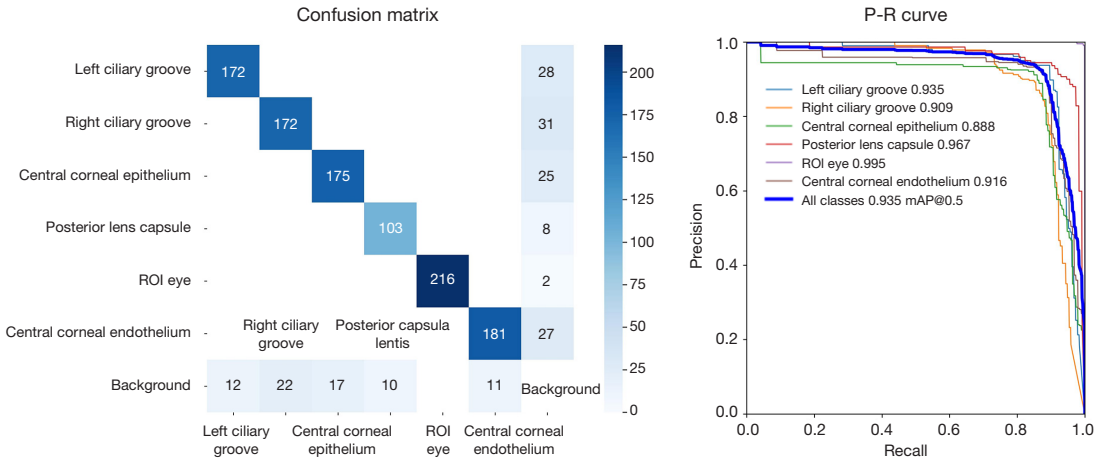


Figure 4 Confusion matrix and P-R curve. mAP, mean average precision; P-R, precision-recall; ROI, region of interest.

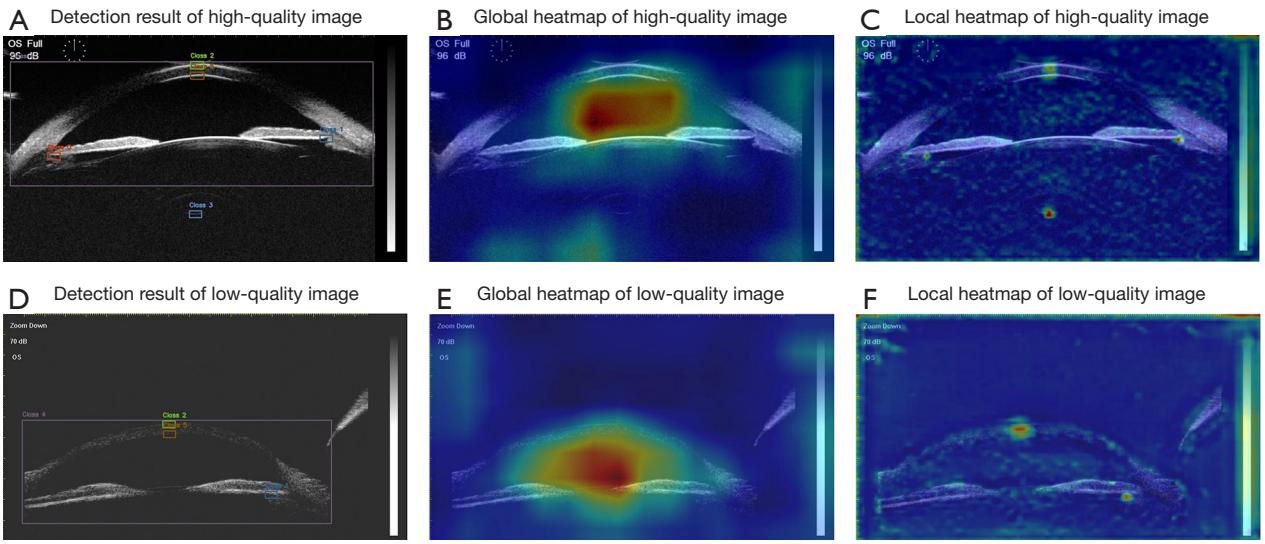


Figure 5 Visualization of heatmaps and detection outcomes for high-quality and low-quality images using deep learning techniques. (A) Detection result of high-quality image. (B) Global heatmap of high-quality image. (C) Local heatmap of high-quality image. (D) Detection result of low-quality image. (E) Global heatmap of low-quality image. (F) Local heatmap of low-quality image.

on the smaller key point areas. This demonstrates that the model effectively concentrates on critical regions during target recognition tasks.

The t-SNE plots generated from both internal and external datasets are shown in *Figure 6*. The internal dataset consists of 1,154 UBM images used for training and testing, whereas the external dataset comprises 892 newly acquired UBM images.

By analyzing the t-SNE plots of the YOLOv8s_DW_FOCUS model on both internal and external datasets, the following key characteristics emerged:

(I) Clear Inter-Class Separation: The feature points of different categories are distinctly separated in the low-dimensional space, demonstrating the model's strong ability to differentiate between various classes. Although some overlap exists in the clustering of the central corneal epithelium and central corneal endothelium, this overlap reflects their close proximity in reality and aligns with their actual spatial relationship, rather than indicating a lack of discriminatory power in the model for these categories.

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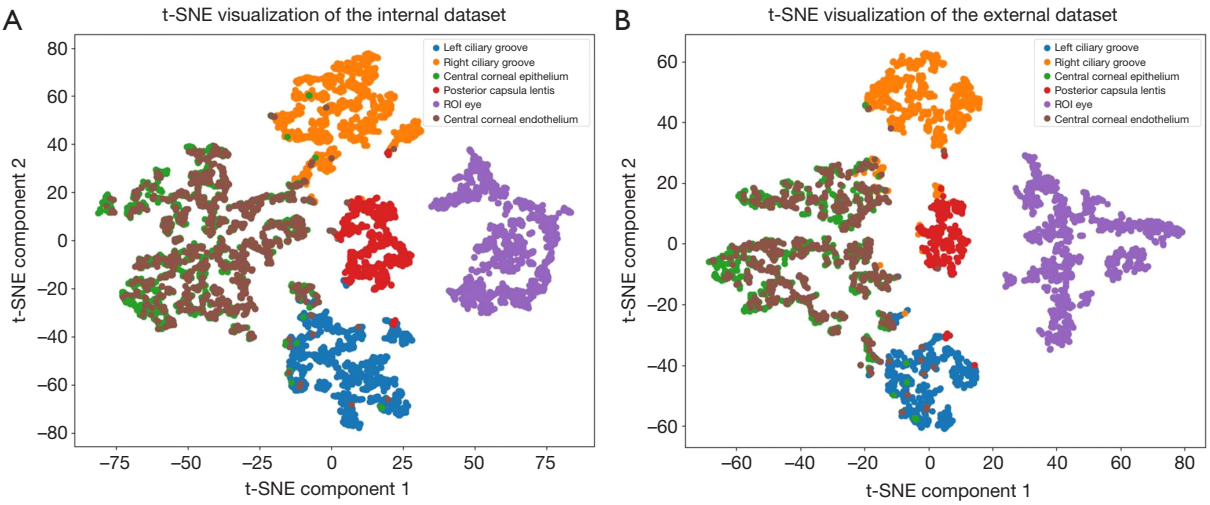


Figure 6 t-SNE plots of internal (A) and external (B) datasets. ROI, region of interest; t-SNE, t-distributed stochastic neighbor embedding.

(II) Compact Intra-Class Clustering: Feature points for each category form relatively tight clusters, indicating the model's high accuracy in recognizing individual categories.

(III) Consistent Distribution Features: The t-SNE plots for both the internal and external datasets exhibit similar distributions, reflecting the model's consistent performance across different datasets and highlighting its robust generalization capability.

These results suggest that the YOLOv8s_DW_FOCUS model excels in detecting targets in anterior segment UBM images, demonstrating superior category discrimination, recognition accuracy, and generalization ability.

After validating the target detection performance of the YOLOv8s_DW_FOCUS model, clinical UBM videos were analyzed using the proposed no-reference quality assessment method, which incorporates the eight previously mentioned quality control indicators. The quality evaluation results are presented in *Figure* 7.

Based on the scoring rules of the eight quality control indicators, the individual scores and total score for each frame are displayed in real-time in the top-right corner of the image. The results show that as the scores decrease, there is a noticeable decline in image quality, making it easily distinguishable. Verification of the pixel-level and geometric features revealed that the DL-based assessment results are generally reasonable and effectively reflect the image quality.

Finally, 174 standard images, selected subjectively by clinical doctors, were input into the no-reference quality assessment method proposed in this study. A comparison between the results of the two quality control methods was performed. *Figure 8* presents the score distribution for these 174 images, along with the reasons for the lost points.

As shown in the figure, the image scores are primarily concentrated in the 7–8 point range, with only a few images exhibiting significant differences in evaluation results. This indicates a high degree of consistency between the no-reference quality assessment model and clinical practice. The lost points are mainly attributed to PCHC and CGFZ, reflecting the fact that when selecting standard images, clinical doctors were not overly strict about whether the image was perfectly centered or whether the left and right ciliary grooves fell within the focal zone. In contrast, the no-reference quality assessment model applies stricter quality control criteria, resulting in higher-quality standard images. Relaxing the standards for PCHC and CGFZ would likely improve the consistency between the two evaluation methods.

Figure 9 presents the evaluation results of the standard images, subjectively selected by ophthalmologists, using the no-reference quality assessment model proposed in this study.

The following insights can be drawn from *Figure 9*: the low score for the 5-point image is attributed to the DL model's failure to detect the posterior lens capsule in this frame, resulting in a score of 0 for PCV, PCHC, and AVA. However, the posterior lens capsule is clearly visible to the human eye within this frame. For the 6-point image, human observation confirms that PCHC and CGFZ meet

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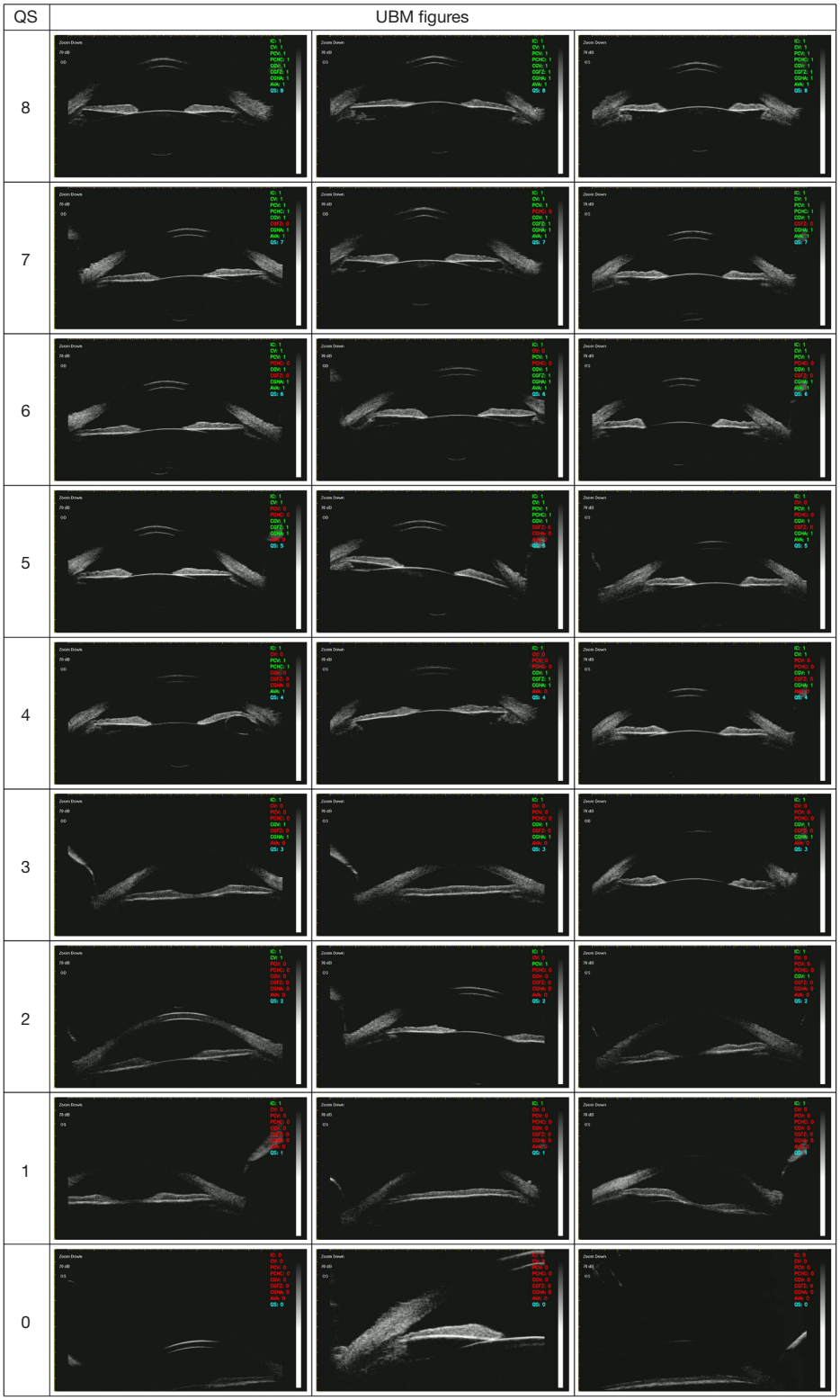


Figure 7 Quality assessment results for UBM images. AVA, Anterior Vertical Alignment; CGFZ, Ciliary Groove Focal Zone; CGHA, Ciliary Groove Horizontal Alignment; CGV, Ciliary Groove Visibility; CV, Cornea Visibility; IC, Image Clarity; PCHC, Posterior Capsule Horizontal Centering; PCV, Posterior Capsule Visibility; QS, quality score; UBM, ultrasound biomicroscopy.

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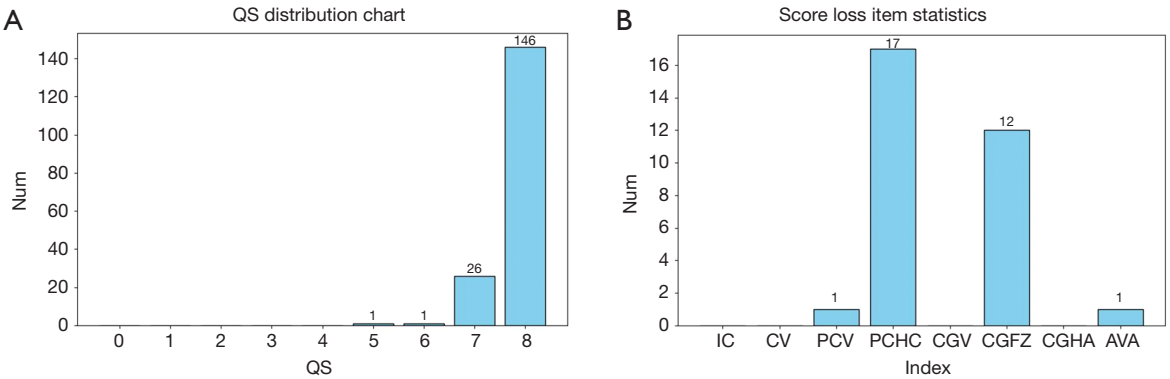


Figure 8 Distribution of scores and statistics of lost points for subjectively selected standard images. (A) QS distribution chart. (B) Score loss item statistics. AVA, Anterior Vertical Alignment; CGFZ, Ciliary Groove Focal Zone; CGHA, Ciliary Groove Horizontal Alignment; CGV, Ciliary Groove Visibility; CV, Cornea Visibility; IC, Image Clarity; PCHC, Posterior Capsule Horizontal Centering; PCV, Posterior Capsule Visibility; QS, quality score.

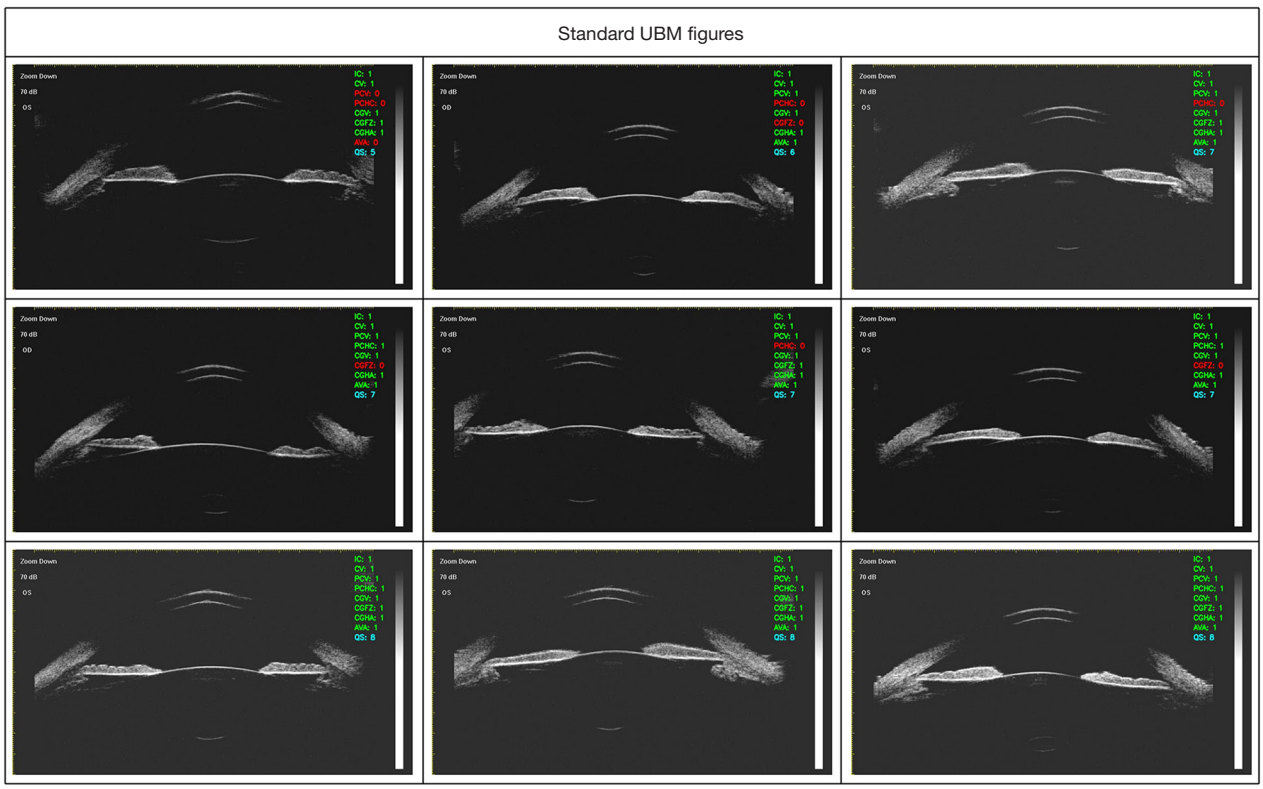


Figure 9 No-reference evaluation results of subjectively selected standard images. AVA, Anterior Vertical Alignment; CGFZ, Ciliary Groove Focal Zone; CGHA, Ciliary Groove Horizontal Alignment; CGV, Ciliary Groove Visibility; CV, Cornea Visibility; IC, Image Clarity; PCHC, Posterior Capsule Horizontal Centering; PCV, Posterior Capsule Visibility; QS, quality score; UBM, ultrasound biomicroscopy.

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the specified requirements, yet the model's object detection results exhibit discrepancies, suggesting that the no-reference quality assessment model may occasionally make errors. In the case of the 7-point images, the posterior lens capsule was not located within the central ±1 mm range, and the left and right ciliary grooves were not fully within the focal zone, which aligns with the actual situation. The 8-point images, having undergone stricter screening, demonstrate higher image quality.

Overall, although the no-reference quality assessment model occasionally makes errors, its evaluation results are highly consistent with clinical practice. This highlights the model's strong reliability and its ability to effectively identify higher-quality standard anterior segment UBM images.

Discussion

DL methods in medical imaging have primarily focused on areas such as disease classification, computer-aided diagnosis, tissue segmentation, image registration, and content retrieval. However, research on using DL for medical IQA remains relatively limited. Existing studies on DL applications in IQA have predominantly concentrated on fields such as fetal cardiac four-chamber views, MRI, and retinal images. To the best of our knowledge, no research has yet been conducted on DL-based quality assessment for anterior segment UBM images.

Currently, the quality assessment of anterior segment UBM images largely relies on traditional subjective evaluation methods, which are time-consuming, labor-intensive, and subject to significant observer variability, making it difficult to maintain consistent evaluation accuracy. This highlights the urgent need for an accurate, reliable, and efficient quality assessment method for anterior segment UBM images in clinical practice, in order to reduce the workload of radiologists.

Furthermore, due to the complexity of human eye structures, individual differences, and the high level of patient cooperation required for UBM image acquisition, obtaining standard anterior segment images is a challenging task. In clinical environments, characterized by complexity and strict diagnostic requirements, clinicians often struggle to quickly and accurately isolate the standard anterior segment UBM sections necessary for parameter measurement and disease diagnosis.

UBM imaging presents several modality-specific challenges: probe misalignment can shift critical anatomical structures away from the image center, diminishing diagnostic value; acoustic shadowing and air bubble artifacts reduce image clarity and obscure essential features; limited acoustic impedance differences among ocular tissues result in low contrast and indistinct anatomical boundaries; and variability in scanning planes introduces spatial inconsistencies that affect reproducibility and quantitative analysis. These factors collectively increase the complexity of quality control in UBM imaging.

Given these challenges, traditional NR-IQA methods such as Blind/Referenceless Image Spatial Quality Evaluator (BRISQUE), Natural Image Quality Evaluator (NIQE), and Blind Image Quality Index (BIQI), primarily developed for natural images, are insufficient for addressing these challenges, as they do not consider domain-specific anatomical structures or spatial relationships crucial for medical interpretation (28-30).

To overcome these limitations, we propose a DL-based NR-IQA method for anterior segment UBM images, integrating structure-aware evaluation and clinical quality control criteria. Our method based on the YOLOv8s_DW_FOCUS model and incorporates eight clinically relevant quality control indicators (IC, CV, PCV, PCHC, CGV, CGFZ, CGHA, and AVA), developed in collaboration with ophthalmic imaging experts. This method ensures a scientifically grounded and objective evaluation standard.

The model accurately identifies detailed anatomical structures by detecting five key landmarks—central corneal epithelium, central corneal endothelium, posterior lens capsule, left and right ciliary grooves—and quantifies their spatial and pixel-level features to assess image quality comprehensively.

Compared to traditional methods such as BRISQUE, NIQE and BIQI, our proposed method offers notable advantages in structural recognition, adaptability, and clinical applicability. By integrating clinical anatomical prior knowledge, the model not only evaluates image sharpness but also determines whether key anatomical structures are sufficiently visible for diagnostic purposes. The eight quality control indicators–covering sharpness, structural visibility, positional accuracy, and geometric consistency–are specifically designed to reflect the practical quality requirements of UBM images in clinical settings. In contrast, traditional methods rely on statistical assumptions derived from natural images and often fail to accommodate the complex textures and low-contrast boundaries typical of medical imaging. Our DL-based approach overcomes these limitations by learning the intrinsic relationships

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between anatomical structures and image quality, thereby demonstrating superior domain specificity, robustness, and generalizability in UBM IQA.

Additionally, a comprehensive analysis of the method's accuracy, focus on key regions, generalization ability, inter-class and out-of-class discriminability, rationality, and consistency of evaluation results confirmed the feasibility of the no-reference quality assessment approach. This method not only enables efficient quality assessment of anterior segment UBM images but also provides reliable and practical technical support for ophthalmic clinical practice.

Compared to existing AI approaches applied to medical IQA, the method proposed in this study offers significant advantages in terms of evaluation comprehensiveness, interpretability, and clinical relevance. Although the model by Piccini *et al.* (18) achieved a high level of agreement with clinical experts, it primarily focused on anatomical structure clarity and coronary artery visualization, resulting in a limited set of evaluation criteria. This narrow scope leads to substantial uncertainty when assessing images of intermediate quality. By contrast, our method combines general pixel-level metrics with ophthalmic anatomical knowledge, incorporating a rich and diverse set of indicators. The use of a 0–8 point grading system allows for a more continuous and interpretable assessment of image quality. In the model created by Esses *et al.* (19), the quality assessment of liver MRI images did not include diagnostic information such as lesion detectability or organ morphology, thereby limiting the model's ability to reflect clinical image utility. Our method addresses this limitation by emphasizing critical anatomical structures such as the posterior lens capsule, ciliary sulcus, and cornea, and by incorporating their spatial location and echo characteristics. This leads to a more robust and comprehensive evaluation, with enhanced resistance to artifacts and noise. Abramovich *et al.* (22) developed a fundus image quality scoring model based on binary classification, which fails to capture the continuous nature of medical image quality. In contrast, our method employs a graded scoring approach, offering clearer distinctions in quality levels and yielding more intuitive and clinically meaningful results. Although Pérez *et al.* (25) achieved accurate assessment of fundus image quality on mobile platforms, their model only provided an overall acceptability score, limiting interpretability. Our approach quantitatively evaluates image quality across multiple dimensions, allowing clinicians to clearly understand both the strengths and limitations of each image. In summary, the proposed method integrates quantitative image quality metrics with domain-specific ophthalmic knowledge. Compared to existing approaches, it provides superior interpretability, visual clarity, and multidimensional assessment capability, demonstrating strong potential for practical clinical deployment.

The target detection method built on the YOLOv8s_DW_FOCUS model demonstrated exceptional performance, with an mAP value of 0.935, recall of 0.898, and precision of 0.925. The model effectively focused on key areas in heatmaps, and t-SNE plots validated its high discriminability between classes and robust generalization performance. These results indicate that the target detection method can effectively and accurately detect the ROI range and the positions of the five key points in anterior segment UBM images, providing a solid foundation for the parameters involved in quality control.

Moreover, this study provides highly discriminative IQA results through the eight key quality control indicators, ensuring the reliability of the parameters. To validate the consistency between the proposed no-reference quality assessment model and subjective assessments by clinicians, 174 standard images, subjectively selected by clinicians, were input into the model. The results showed that 172 images scored 7 or 8, indicating a high level of agreement between the two evaluation methods. An analysis of the reasons for score discrepancies revealed two contributing factors: first, a small number of images scored lower due to errors in target detection; second, the stricter evaluation standards of the no-reference quality assessment model resulted in a more rigorous selection process for images, leading to discrepancies with subjective evaluations. These findings further demonstrate that the proposed method can not only quickly extract standard cuts from anterior segment UBM images but also efficiently and accurately evaluate image quality, significantly enhancing the efficiency of imaging specialists. Although the method shows significant potential, its clinical efficacy requires further validation through practical application and clinical trials.

From a clinical application perspective, the proposed model demonstrates strong feasibility for deployment. The model was trained and validated on a single NVIDIA RTX 3090Ti GPU, achieving an inference speed of 41.44 FPS—far exceeding the 3–5 fps typically required for anterior segment UBM imaging, thereby fully meeting real-time processing demands. The YOLO architecture inherently offers efficient data handling and low hardware dependency. By incorporating DWConv and the FOCUS module, we constructed a lightweight version—YOLOv8s_

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DW_FOCUS—which effectively reduces computational overhead. Given the moderate resolution (1,024×576) and frame rate of UBM images, the model runs smoothly on standard hospital workstations or even devices equipped with lightweight GPUs, making it particularly suitable for resource-limited clinical environments.

The model can be seamlessly integrated as a plug-in into existing UBM imaging systems, enabling real-time image quality scoring and feedback with minimal user intervention. This allows clinicians to immediately assess image diagnostic suitability during acquisition, facilitating the rapid collection of high-quality images. Such design offers notable advantages in terms of system compatibility, scalability, and clinical practicality.

Nevertheless, several challenges remain for broader clinical implementation. These include integration with hospital information systems, compliance with data privacy regulations, and the need for model interpretability expected by healthcare professionals. Therefore, in future work, we will further optimize the human–computer interaction interface of the model, enhance its interpretability, and conduct clinical validation in complex real-world settings to facilitate its clinical implementation.

Beyond improving workflow efficiency, the proposed system also holds significant potential for enhancing patient outcomes. On the one hand, standardized quality control can help to reduce misdiagnosis or missed diagnoses caused by suboptimal image quality. On the other hand, improving consistency in image follow-up facilitates the assessment of disease progression and evaluation of treatment efficacy. Therefore, we believe that this system holds great promise for improving diagnostic accuracy in clinical practice and optimizing the allocation of medical resources.

This study has some limitations. First, the data source was somewhat limited: all UBM images were collected using the MD-300L model, with fixed image resolution and format, and the data were only from patients at Tianjin Eye Hospital. The sample does not include data from multiple countries or diverse ethnic groups, which somewhat restricts the generalizability of the no-reference quality assessment model. To address this, we plan to actively expand collaborations with other medical centers and UBM device manufacturers in order to incorporate more diverse data sources, imaging modalities, and pathological cases. Such efforts will help improve the robustness and adaptability of the model across different devices and clinical conditions. Moreover, we highly value the suggestion to conduct subgroup analyses involving anterior segment diseases such as glaucoma and cataracts. Future research will aim to incorporate a broader range of pathological data to explore the model's performance and reliability in these clinically relevant subgroups.

Second, the target detection model still has some errors in the detection results. Future work should focus on optimizing the model structure and increasing the dataset size to further improve detection accuracy. Additionally, the recognition of the eight quality control indicators proposed in this study requires further enhancement. These indicators were developed in collaboration with ophthalmic imaging specialists at Tianjin Eye Hospital, and their standards need to be verified and refined through additional research and expert validation. This will help to further improve the no-reference quality assessment system for anterior segment UBM images.

Conclusions

Through accuracy comparison tests of the DL model, verification of key region focus, evaluation of inter-class and out-of-class performance, as well as analyses of the accuracy, rationality, and consistency of the no-reference quality assessment method, the proposed DL-based no-reference quality assessment method for anterior segment UBM images demonstrates exceptional performance. This method facilitates accurate, objective, and real-time IQA. The results indicate that the assessment method aligns closely with subjective evaluations by ophthalmic imaging specialists, while applying more stringent image quality control standards. It can effectively assist clinicians in quickly extracting standard anterior segment UBM cross-sectional views and accurately selecting high-quality images, significantly improving the efficiency of imaging specialists and offering substantial clinical value for widespread adoption.

Although the method shows great potential, its practical performance requires further validation in the complex and dynamic clinical environment. With continuous optimization and enhancement, it is expected to be gradually adapted for broader clinical use.

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None.

Footnote

Reporting Checklist: The authors have completed the CLEAR

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reporting checklist. Available at https://qims.amegroups.com/article/view/10.21037/qims-2025-592/rc

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Conflicts of Interest: All authors have completed the ICMJE uniform disclosure form (available at https://qims.amegroups.com/article/view/10.21037/qims-2025-592/coif). The authors have no conflicts of interest to declare.

Ethical Statement: The authors are accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved. The study was conducted in accordance with the Declaration of Helsinki and its subsequent amendments. Ethical approval was approved by the Medical Ethics Committee of Tianjin Eye Hospital (Approval No. KY-202417). Since this was a retrospective study utilizing de-identified UBM images, the requirement for informed consent was waived.

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